Serife Damla Konur

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Experience

ImFusion GmbH June 2025 – Current
Computer Vision Engineer - Work Study *Munich, Germany*

- Responsible for deploying and testing state-of-the-art computer vision models using Python and C++.
- Involved in research projects on 2D and 3D point tracking, including evaluating and extending recent models such as CoTracker3.

Brainlab AG November 2023 – February 2025
Working Student *Munich, Germany*

- Created analysis dashboards for customers using mainly Python Dash and scikit-learn.

Tradeware Technology August 2022 – October 2023
Software Engineer *Istanbul, Turkey*

- Designed and developed high-frequency, multi-threaded trading platforms with low latency using C++.

Artificial Intelligence and Intelligent Systems (AI^{2}S) Laboratory September 2020 – July 2021
Undergraduate Research Fellow *Istanbul, Turkey*

- Funded by Scientific and Technological Research Council of Turkey (TUBITAK).
- The research interests were mainly advancing vision based lane detection algorithms and implementing different controller structures for lateral and longitudinal control of MIT Racecar.

Education

Technical University of Munich October 2023 - Current
MSc. Informatics *Munich, Germany*

- Specialization : Computer Graphics and Vision
- Thesis : FloorPlan Based 3D Scene Generation

Istanbul Technical University 2018 – 2022
BSc. Computer Engineering (Double Major) - Grade: 1.8 *Istanbul, Turkey*

- Thesis : Structuring and Reporting Twitter Posts of Twitter Influencers About Istanbul Stock Market

Istanbul Technical University 2016 – 2021
BSc. Control and Automation Engineering - Grade: 1.6 *Istanbul, Turkey*

- Thesis : Vision Based Trajectory Control of the Mini Autonomous Car

Projects

Generative Model of 3D Heads

- Participated in the practical course: 3D Scanning & Spatial Learning. Trained an auto-decoder to learn the latent distribution of human heads using Cafca Dataset. The learned prior is used for 3D head reconstruction from a few input images by using **Python**, **Pytorch**, **3D Gaussian Splatting**.

Semantic 3D Gaussian Splatting

- Master project Extended the original 3D Gaussian Splatting pipeline with per-Gaussian semantic features by implementing custom CUDA forward/backward rasterization kernels, joint RGB+semantic optimization for obtaining consistent 3D semantic representation. **Python**, **PyTorch**, **CUDA**, **C++**, **3D Gaussian Splatting**.

Multi-View Fusion for 3D-Complete Head Avatars (Guided Research)

- Fine-tuned a pretrained single-view avatar model for a two-view setup within a 10 GB budget, reducing ambiguity and improving 3D completeness while preserving FLAME-controlled animation. Added lightweight multi-view fusion and updated only small heads, improving novel-view consistency on multi-view data.

Technical Skills

Languages: C/C++, Python
Frameworks and Libraries:Pytorch, Tensorflow, OpenCV, Eigen, Ceres, Pthread, Gtest, Pytest, Flask
Skills: Deep Learning, Computer Vision, 3D Geometry, 3D Gaussian Splatting, NeRF, Linux, CMake, Multi-Threading, Bash-Scripting, Git, Docker